

Type: Practical (Minor) B.Sc(ECS)-II (Semester III) Course Title: Practical based on DSC2-3 (Paper Code:)	
Total Lectures: 30 Hrs. Contact Hrs. (P) : 2	
University Evaluation: 15 Marks Internal Evaluation: 10 Marks	
1.	Write Python scripts to illustrate the concepts of statistical population and sample. Generate a random sample from a given population and calculate sample statistics (mean, median, mode, variance, etc.), then compare them with population parameters.
2.	Implement Python functions to simulate different sampling methods and analyze their effectiveness using statistical metrics.
3.	Develop Python functions to process and condense data. Convert raw data into variables (discrete and continuous), create frequency distributions (ungrouped and grouped), and calculate cumulative frequency distributions.
4.	Develop Python scripts to process raw data into frequency distributions and plot histograms and ogives. Analyze graphical representations to interpret data trends.
5.	Solve numerical problems related to data condensation and graphical methods using Python. Compute frequencies and cumulative frequencies, and plot histograms for given datasets. Analyze and interpret graphical representations to conclude data distributions.
6.	Write Python functions to compute measures of central tendency: I. Arithmetic Mean (AM) for ungrouped and grouped data II. Median for ungrouped and grouped data using formulas and graphical methods III. Mode for ungrouped and grouped data, including handling multimodal distributions
7.	Write Python scripts to compute relative measures of dispersion: I. Coefficient of Range and Coefficient of Variation for datasets with different scales and origins II. Discuss the impact of scale and origin changes on these measures
8.	Write Python script to plot scatter diagram when $r = -1$, $r = 0$ and $r = +1$.
9.	Develop Python functions to compute Karl Pearson's coefficient of correlation. Calculate the correlation coefficient (r) for given datasets and validate calculations against theoretical limits ($-1 \leq r \leq 1$).
10.	Develop Python scripts to perform regression analysis for given datasets. Interpret regression results.